

Ch 2 Database System Concepts and Architecture

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Lecture 1

Database System Architecture

- Database architecture refers to the overall design and structure of a Database Management System (DBMS).
- It defines how data is stored, managed, and accessed by users and applications.
- In the past, DBMSs were monolithic systems where all functions were integrated into one software package.
- Modern DBMSs use Client/Server, Distributed, and Cloud architectures.

Client/Server Architecture

A **Client/Server architecture** divides the system into two parts:

Client: The client interacts with users.

Functions:

- User Interface (UI)
- Forms
- Applications
- Mobile Apps

Server: The server manages the database.

Functions:

- Data Storage
- Query Processing
- Security



Example: Student Registration System

- Student fills registration form (Client)
- Database stores information (Server)

Data Model

- A Data Model is a collection of concepts used to describe **Data, Relationships, Constraints**.
- It provides a way to organize and represent data.

Example: Student and Course relationship



Schema and Instance

Schema

Schema is the structure or blueprint of a database.

Example: **STUDENT(SID, Name, Major)**

Schema **rarely** changes.

Instance

Instance is the actual data stored in the database at a particular time.

Example

SID	Name
S1	Aung Aung
S2	Su Su

Three-Schema Architecture

The three-Schema Architecture divides the database into three levels:

- External Level
- Conceptual Level
- Internal Level

Purpose:

- Data Independence
- Security
- Multiple User Views

External Level: The External Level shows only the data required by a particular user.

Example: **Teacher View**

StudentID	Name	Grade
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- The **internal level** has an **internal schema**, which describes the physical storage structure of the database.
- The internal schema uses a physical data model and describes the complete details of data storage and access paths for the database.

Note: **Internal Level:** Describes how data is physically stored.

Example: **Files, Indexes, Storage Blocks**

- The **conceptual level** has a **conceptual schema**, which describes the structure of the whole database for a community of users.
- The conceptual schema hides the details of physical storage structures and concentrates on describing entities, data types, relationships, user operations, and constraints.
- Usually, a representational data model is used to describe the conceptual schema when a database system is implemented.
- This *implementation conceptual schema* is often based on a *conceptual schema design* in a high-level data model.

Note: **Conceptual Level:** Represents the entire logical structure of the database.

Example: **STUDENT, COURSE, ENROLLMENT**

- The **external** or **view level** includes a number of **external schemas** or **user views**.
- Each external schema describes the part of the database that a particular user group is interested in and hides the rest of the database from that user group.
- As in the previous level, each external schema is typically implemented using a representational data model, possibly based on an external schema design in a high-level conceptual data model.

Data Independence

Data Independence means changes at one level do not affect higher levels.

1. Logical Data Independence

Changes in database structure **do not affect user views**.

Example: Adding Email column

Student(SID, Name, Email)

2. Physical Data Independence

Changes in storage do not affect database structure.

Example: Hard Disk → SSD

Applications continue working.

2.3 Database Languages and Interfaces

2.3.1 DBMS Languages

DDL (Data Definition Language): Used to define database structure.

It specifies Tables, Attributes, Relationships and Constraints

Commands: **CREATE, ALTER, DROP**

Example

```
CREATE TABLE Student(  
  StudentID INT PRIMARY KEY,  
  Name VARCHAR(50),  
  Major VARCHAR(30)  
);
```

This defines the logical structure.

After the **database schema** has been created (using DDL) and the **database has been filled with data**, users need a way to work with that data.

The **Data Manipulation Language (DML)** is the language used to:

- Retrieve data
- Insert new data
- Update existing data
- Delete data

Example:

```
INSERT INTO Student  
VALUES (1,'Aung Aung','CS');
```

Commands: **INSERT, UPDATE, DELETE**

Retrieve Data (SELECT)

Retrieve means **getting data from the database**.

```
SELECT *
```

```
FROM Student;
```

2.3.2 DBMS Interfaces

What is a DBMS Interface?

A **DBMS Interface** is the medium through which users communicate with the database.

Different users need **different interfaces depending on their knowledge and tasks**.

Types of DBMS Interfaces

1. Menu-Based Interface

A menu-based interface provides users with lists of options (menus).

Users simply choose options instead of typing SQL commands.

Example:

- ATM machine menus
- Website navigation menus
- Online shopping categories

Main Menu

1. Search Student
2. Add Student
3. Delete Student
4. Exit

Advantages

- Easy for beginners
- No need to remember SQL syntax

2. Apps for Mobile Devices

Many organizations provide mobile apps so users can access their databases.

Examples:

- Banking apps
- Hotel reservation apps
- Insurance apps

3. Forms-Based Interface

Users enter information into forms.

The DBMS processes the entered data.

Advantages

- Very easy for non-technical users
- Reduces typing errors

4. Graphical User Interface (GUI)

GUI allows users to interact with the database through graphical elements such as:

- Windows
- Icons
- Buttons
- Diagrams

5. Natural Language Interface

Users ask questions in ordinary language.

The system translates the question into a database query.

Example:

Show all Computer Science students.

Advantages

- No SQL knowledge required
- Easy for everyone

7. Speech Input and Output

Users speak instead of typing.

The system converts speech into queries and can also answer using speech.

Example

"What is my account balance?"

6. Keyword-Based Database Search

Works like a search engine.

Users type keywords instead of SQL.

Example: type **Aung Aung**

The system searches matching records.

8. Interfaces for Parametric Users

Parametric users **perform the same tasks repeatedly.**

The interface **provides shortcut buttons or function keys.**

Examples:

- Bank tellers
- Cashiers
- Ticket counter staff

9. Interfaces for DBA

The Database Administrator (DBA) uses special interfaces with administrative privileges.

Examples:

- Create user accounts
- Grant permissions
- Modify schema
- Backup database
- Reorganize storage

2.4 The Database System Environment

What is Database System Environment?

A **Database System Environment** is the complete environment in which a DBMS operates.

It includes:

- Users
- DBMS software modules
- Operating System (OS)
- Database
- System Catalog (Data Dictionary)

Top part is User Side.

Bottom part is DBMS Internal Processing.

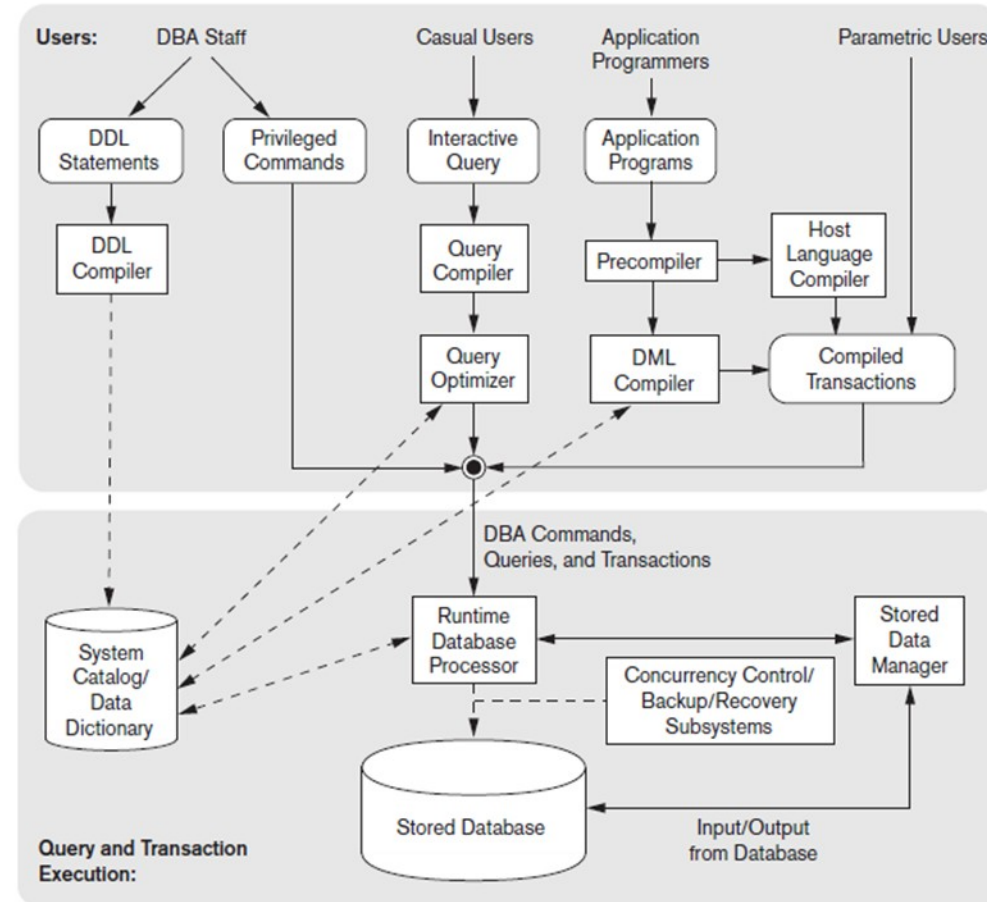


Figure 2.3
Component modules of a DBMS and their interactions.

DBA → DDL Compiler → System Catalog

Casual User → Query Compiler → Query Optimizer → Runtime Database Processor

Application Programmer → Precompiler → DML Compiler + Host Language Compiler → Compiled

Transaction → Runtime Database Processor

Runtime Database Processor → Stored Data Manager → Stored Database